

Drishti-Kavach: Advancing India's Railway Security System with Spatial Clearance Perception Engine for Next-Generation Railway Safety

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Abstract—Traditional railway safety mechanisms—including fixed block track circuits, axle counters, and transponder-based Train Collision Avoidance Systems (TCAS / Kavach)—rely strictly on block-occupancy signals and cab-signaling telemetry. While effective against train-to-train collisions, these systems lack forward optical perception and are fundamentally blind to non-signaled track intrusions, level-crossing vehicular blockades, fallen trees, landslides, and deliberate track sabotage (e.g., steel rods, placed rail sections, and flammable canisters). In this paper, we present Drishti-Kavach, an edge-deployable, dual-spectrum optical Automatic Train Protection (ATP) perception engine engineered for real-time locomotive forward lookahead. Drishti-Kavach decouples visual perception into a lightweight bilateral semantic segmentation network (BiSeNetV2, $\sim 3.49\text{M}$ parameters) for drivable ballast gauge and steel rail ribbon extraction, paired with a high-resolution (1024×1024) YOLO11m detector for micro-sabotage and physical hazard localization. To resolve 24/7 environmental degradation, an atmospheric optical enhancer dynamically couples Dark Channel Prior (DCP) transmission estimation with Luminance-domain CLAHE, operating in tandem with an active 850nm Near-Infrared (NIR) headlamp modeling framework. Rather than employing black-box collision classification, a deterministic Vector Spatial Clearance Engine applies Shapely polygon unions, base ground footprint projections, and dynamic lateral safety buffering to classify threats into a 3-tier hierarchy (CRITICAL, WARNING, CLEAR). Evaluated across a multimodal benchmark combining RailSem19 and Indian UAV-RSOD datasets, Drishti-Kavach achieves 86.02% Universal mIoU (90.08% Track Bed IoU) and 90.90% AP@50 on deliberate track sabotage (*IronRod*), while operating at >50 FPS on edge accelerators with sub-20ms total latency.

Index Terms—Automatic Train Protection (ATP), Intelligent Transportation Systems (ITS), Semantic Segmentation, Obstacle Detection, BiSeNetV2, YOLO11m, Active Near-Infrared Vision, Spatial Clearance Geometry, Railway Sabotage Prevention.

I. INTRODUCTION

Modern railway networks represent the critical infrastructure backbone of national economies. On high-density rail

corridors such as the Indian Railways network, safety has historically depended on automated signaling infrastructures, fixed block circuits, and radio-frequency transponders such as **Kavach** (TCAS - Train Collision Avoidance System). While Kavach enforces Signal Passed at Danger (SPAD) prevention and locomotive-to-locomotive head-on collision avoidance, recent surveys on railway safety [1], [2] underscore an intrinsic structural limitation: *conventional transponders cannot perceive physical track blockades ahead of the train*.

Physical derailments and catastrophic railway collisions frequently occur due to un-signaled obstructions lying outside the capability envelope of track circuits [1]:

- 1) **Deliberate Sabotage:** Foreign steel girders, severed rail pieces, or fishplates placed across running rails to induce derailment.
- 2) **Level-Crossing Gate Entrapment:** Passenger road vehicles, trucks, and buses stalled within the railway clearance envelope.
- 3) **Geological and Washout Hazards:** Boulders from mountain rockfalls, washed-out ballast, and storm-felled tree limbs.
- 4) **Human and Wildlife Trespassing:** Track maintenance gangs, pedestrians, and livestock crossing blind curves.
- 5) **Atmospheric Blindness:** Severe winter radiation fog, monsoon precipitation, and night darkness that impair locomotive drivers' vision beyond safe emergency braking distances.

To bridge this critical safety gap, we introduce **Drishti-Kavach**, an optical Automatic Train Protection perception engine. Our primary technical contributions are:

- **Decoupled Dual-Engine Architecture:** We eliminate gradient conflict between dense boundary segmentation and small-object detection by separating track bed parsing

(BiSeNetV2) from micro-sabotage detection (YOLO11m at 1024×1024).

- **Universal Dual Training Protocol:** We introduce a multimodal dataset synthesis strategy integrating first-person locomotive cab feeds (RailSem19) with aerial/infrastructure feeds (UAV-RSOD V1), regularized with physics-based Active 850nm NIR Night Vision synthesis to prevent catastrophic forgetting.
- **Deterministic Vector Spatial Clearance Engine:** Moving away from brittle black-box classifiers, we employ Shapely vector polygon geometry, bottom 25% ground footprint projection, and dynamic lateral buffers ($\delta = 65\text{px}$) to compute verifiable ATP intervention triggers.
- **Atmospheric Optical Enhancer:** An adaptive multi-mode module that scores atmospheric fog density via Dark Channel Prior (DCP) statistics and executes localized LAB Luminance CLAHE for haze and monsoon rain suppression.

II. RELATED WORK

A. Railway Intrusion Detection & Perception Frameworks

Comprehensive reviews by Xu *et al.* [1] emphasize that vision-based onboard obstacle intrusion detection systems (OIDSs) are paramount for autonomous train safety. While early research relied on stationary trackside surveillance, on-board vision enables continuous corridor coverage. Chen *et al.* [2] proposed multitask perception learning for real-time obstacle detection across railway sectors, demonstrating that multi-task pipelines must maintain rigorous real-time performance without catastrophic accuracy loss.

B. Deep Learning for Railway Obstacle and Sabotage Detection

To overcome the limitations of general object detectors on railway tracks, Wang and Du [3] introduced **YOLO-Rail**, utilizing ConvNeXtV2 and feature fusion to detect obstacles on rails. Li *et al.* [5] proposed an improved RT-DETR framework tailored for railway obstacle detection, demonstrating the power of real-time transformer backbones. Eisentraut *et al.* [8] investigated domain-specific feature extraction for railway obstacle classification, proving that localized physical context is essential to prevent false alarms. In 3D point cloud perception, Lian *et al.* [7] introduced RAE3D, while Liu *et al.* [6] introduced RailBEV for bird's-eye-view representation.

C. Semantic Track Segmentation and Clearance Geometry

Accurate delineation of running rails and drivable ballast is required to define the train clearance envelope. Song *et al.* [9] proposed attention-aggregated semantic segmentation for foreign object intrusion, showing that high-precision boundary extraction reduces spatial ambiguity. Guo *et al.* [4] introduced **Track2Net**, a fast lightweight keypoint alignment model for rail line tracking. In dynamic clearance modeling, Wu and Li [10] demonstrated multi-sensor dynamic obstacle avoidance and path clearance reasoning.

III. SYSTEM ARCHITECTURE & METHODOLOGY

The overall architecture of Drishti-Kavach is illustrated in Fig. 1.

A. Multi-Spectral Active 850nm Near-Infrared (NIR) Modeling

To guarantee 24/7 operational capability, we model the physics of locomotive-mounted active 850nm infrared illuminators. The synthetic NIR conversion pipeline applies spectral CMOS sensitivity weighting, conical spotlight attenuation, tone-mapping, and sensor noise:

$$\text{Mono}(x, y) = 0.18B(x, y) + 0.47G(x, y) + 0.35R(x, y) \quad (1)$$

The conical illuminator is modeled with radial intensity decay centered at the track vanishing axis ($x_0 = W/2, y_0 = 0.60H$):

$$d^2(x, y) = \frac{(x - x_0)^2}{(0.58W)^2} + \frac{(y - y_0)^2}{(0.48H)^2} \quad (2)$$

$$I_{\text{spot}}(x, y) = \exp(-1.4 \cdot d^2(x, y)) \quad (3)$$

$$I_{\text{NIR}}(x, y) = 0.12 + 0.88 \cdot I_{\text{spot}}(x, y) \quad (4)$$

$$\text{Frame}_{\text{NIR}}(x, y) = \text{clip}\left(\left(\frac{\text{Mono} \cdot I_{\text{NIR}}}{255}\right)^{1.15} \cdot 255 + \mathcal{N}(0, 6.0), 0, 255\right) \quad (5)$$

B. Atmospheric Optical Enhancer

The optical enhancement module monitors atmospheric transmission in real time. It computes an **Airlight Fog Density Score** $S_{\text{fog}} \in [0, 1]$ combining luminance standard deviation σ_{gray} and dark channel intensity μ_{dark} :

$$S_{\text{contrast}} = \text{clip}\left(\frac{55.0 - \sigma_{\text{gray}}}{40.0}, 0, 1\right) \quad (6)$$

$$S_{\text{airlight}} = \text{clip}\left(\frac{\mu_{\text{dark}} - 60.0}{100.0}, 0, 1\right) \quad (7)$$

$$S_{\text{fog}} = 0.60 \cdot S_{\text{airlight}} + 0.40 \cdot S_{\text{contrast}} \quad (8)$$

When $S_{\text{fog}} > 0.60$, the system executes Dark Channel Prior (DCP) inversion:

$$J(x) = \frac{I(x) - A}{\max(t(x), t_0)} + A \quad (9)$$

When $0.40 < S_{\text{fog}} \leq 0.60$, it applies Luminance CLAHE in the LAB color space with clip limit 3.0 and grid size (8, 8).

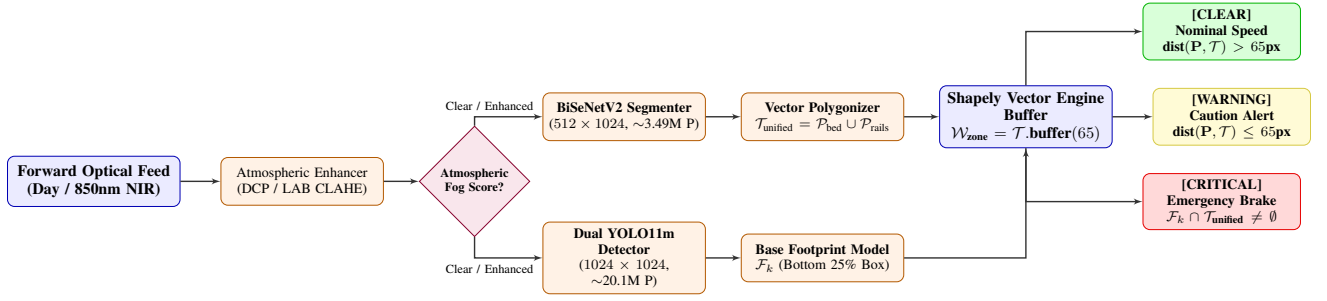


Fig. 1. End-to-End Functional Architecture Flowchart of the Drishti-Kavach Decoupled Optical Automatic Train Protection Engine.

C. Track Bed & Rail Ribbon Segmenter: BiSeNetV2

BiSeNetV2 handles high-speed semantic track segmentation across three canonical classes: 0: *Background*, 1: *Track_Bed*, and 2: *Rail_Lines* [4], [9].

- 1) **Detail Branch (Spatial Pathway):** Uses 3 stages of shallow convolutions down to $1/8\times$ scale with 128 feature channels to retain sharp rail boundaries and ballast textures.
- 2) **Semantic Branch (Context Pathway):** Employs a *StemBlock* ($1/4\times$) and Gather-and-Expansion (*GELayer*, $e = 6$) inverted bottleneck blocks down to $1/32\times$, terminated by a Context Embedding block (*CEBlock*) with Global Average Pooling (GAP).
- 3) **Bilateral Guided Aggregation (BGA):** Fuses spatial details and semantic cues via bidirectional depthwise separable attention gates:

$$\text{Path}_1 = \text{Detail}_{\text{DW}} \odot \sigma(\text{Semantic}_{\text{Up}}) \quad (10)$$

$$\text{Path}_2 = \text{Interp}(\text{Semantic}_{\text{Conv}} \odot \sigma(\text{Detail}_{\text{Down}})) \quad (11)$$

$$\text{Output}_{\text{BGA}} = \text{Conv}_{3\times 3}(\text{Path}_1 + \text{Path}_2) \quad (12)$$

- 4) **Booster Training vs. Zero-Cost Inference:** 4 auxiliary booster heads inject intermediate supervision gradients during training. These heads are pruned prior to ONNX export, ensuring **0 FLOPs overhead** at runtime ($\sim 3.49\text{M}$ parameters, 13.3 MB binary).

D. Dual-Layer Railway Sabotage & Obstacle Detector

Operating natively at 1024×1024 resolution, the detection subsystem runs two cooperating layers [2], [3]:

- **Layer 1 (Custom 8-Class Railway Hazard Engine):** Fine-tuned specifically for *Person*, *Car*, *Truck*, *Branch*, *IronRod*, *Boulder*, *Barrel*, and *Jerrycan*. Trains and locomotives are strictly excluded to prevent false triggers from parallel line traffic.
- **Layer 2 (Complementary Foundation Baseline):** Incorporates 19 enabled COCO foundation classes covering livestock/wildlife, auxiliary transport, and unattended luggage.
- **IoU-Based Fusion:** Merges predictions using non-maximum suppression ($\text{IoU}_{\text{thresh}} = 0.45$), where custom sabotage detections take operational precedence.

IV. VECTOR SPATIAL CLEARANCE & ATP HAZARD ENGINE

Perception bounding boxes and raster masks cannot directly command train braking without spatial context. Drishti-Kavach translates 2D bounding boxes and segmented masks into mathematical vector geometry via the Shapely engine [6], [10].

A. Bottom 25% Ground Contact Footprint Model

To eliminate false overlaps caused by perspective distortion (where an object's head or upper profile projects onto distant background tracks), we model the physical ground contact footprint $\mathcal{F}_k$:

$$\mathcal{B}_k = [x_1, y_1, x_2, y_2], \quad h_k = y_2 - y_1 \quad (13)$$

$$y_{\text{foot}} = y_2 - 0.25 \cdot h_k \quad (14)$$

$$\mathcal{F}_k = \text{Polygon}([(x_1, y_{\text{foot}}), (x_2, y_{\text{foot}}), (x_2, y_2), (x_1, y_2)]) \quad (15)$$

$$\mathbf{P}_{\text{anchor}} = \left(\frac{x_1 + x_2}{2}, y_2 \right) \quad (16)$$

B. Unified Track Union & Dynamic Clearance Buffering

OpenCV contour extraction polygonizes segmented masks into track bed polygons $\mathcal{P}_{\text{bed}}$ and rail line polygons $\mathcal{P}_{\text{rails}}$. A unary union forms the master reference geometry:

$$\mathcal{T}_{\text{unified}} = \text{unary_union}(\mathcal{P}_{\text{bed}} \cup \mathcal{P}_{\text{rails}}) \quad (17)$$

A dynamic lateral safety envelope is constructed with dilation parameter $\delta = 65$ pixels:

$$\mathcal{W}_{\text{zone}} = \mathcal{T}_{\text{unified}}.\text{buffer}(\delta) \quad (18)$$

C. Three-Tier Threat Classification (ATP Braking Rules)

- 1) **[CRITICAL] Threat (In-Track Obstacle $\rightarrow$ ATP Emergency Brake):**

$$\mathcal{F}_k \cap \mathcal{T}_{\text{unified}} \neq \emptyset \quad \vee \quad \mathbf{P}_{\text{anchor}} \in \mathcal{T}_{\text{unified}} \quad (19)$$

Action: Triggers emergency brake line via locomotive TCAS sub-rack interface and paints target bounding box in vivid red.

- 2) **[WARNING] Threat (Near-Track Infringement $\rightarrow$ Driver HUD Caution):**

$$(\mathcal{F}_k \cap \mathcal{W}_{\text{zone}} \neq \emptyset \vee \mathbf{P}_{\text{anchor}} \in \mathcal{W}_{\text{zone}}) \wedge \mathcal{F}_k \cap \mathcal{T}_{\text{unified}} = \emptyset \quad (20)$$

Action: Computes Euclidean track distance $d_k = \text{dist}(\mathbf{P}_{\text{anchor}}, \mathcal{T}_{\text{unified}}) \leq 65\text{px}$ and alerts the driver HUD for speed curtailment.

3) [CLEAR] / SAFE State (Off-Track Object $\rightarrow$ Nominal Line Speed):

$$\text{dist}(\mathbf{P}_{\text{anchor}}, \mathcal{T}_{\text{unified}}) > \delta \quad (21)$$

Action: Validates obstacle clearance (e.g., platform passengers or parallel road vehicles); no brake intervention.

V. EXPERIMENTAL SETUP & DATASETS

A. Curated Datasets

- 1) *dataset_seg/* (RailSem19 Cab View): 8,500 cab-view frames remapped into 3-class canonical format (*Background*, *Track_Bed*, *Rail_Lines*) with 50/50 Day/Active NIR Night pairs.
- 2) *dataset_uav_v1/* (UAV-RSOD Infrastructure): High-angle drone and gantry imagery providing structural angle regularization.
- 3) *dataset_det/* (Unified 8-Class Hazard Dataset): Standardized YOLO bounding boxes combining UAV-RSOD V2 sabotage targets and RailSem19 crossing vehicles.

TABLE I
CURATED DATASETS SPECIFICATION SUMMARY

Curated Directory	Source Benchmark	Output Format	Target Model
<i>dataset_seg/</i>	RailSem19 Cab View	3-Class 8-bit PNG	BiSeNetV2
<i>dataset_uav_v1/</i>	UAV-RSOD V1 Infra	3-Class 8-bit PNG	BiSeNetV2
<i>dataset_det/</i>	UAV-RSOD V2 + RailSem	8-Class YOLO TXT	YOLO11m

B. Training Hyperparameters

- **Universal BiSeNetV2:** Trained using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 1×10^{-4}) with polynomial learning rate schedule (initial LR 5×10^{-4} , power 0.9) over 40 epochs on dual NVIDIA T4/P100 GPUs.
- **YOLO11m Detector:** Trained at native 1024×1024 resolution for 50 epochs using Task-Aligned Assigner (TAL), Mosaic augmentations, and CIoU + DFL loss [3], [8].

VI. EXPERIMENTAL RESULTS & BENCHMARKS

A. Semantic Track Segmentation Performance (Universal BiSeNetV2)

The segmentation engine achieves **86.02% Universal mIoU** and **98.86% Pixel Accuracy** across multi-angle validation sets:

$$\text{Mean IoU (mIoU)} = \frac{1}{C} \sum_{c=0}^{C-1} \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} = \mathbf{86.02\%} \quad (22)$$

TABLE II
SEMANTIC SEGMENTATION PERFORMANCE BREAKDOWN

Semantic Class	IoU (%)	Recall (%)	Precision (%)	Dice / F1 (%)
Track Bed (Ballast)	90.08	94.21	95.34	94.78
Rail Lines (Rails)	69.11	76.50	87.80	81.73
Background / Sky	98.86	99.30	99.55	99.43
Global Summary	86.02	90.00	94.23	91.98

B. Cross-Perspective & Multi-Spectral Invariance

To verify that Universal Dual Training eliminated catastrophic forgetting, accuracy was evaluated across disparate domain splits (Table III).

- **Cross-Domain Stability:** The accuracy delta between cab-view (86.45%) and aerial drone perspective (85.78%) is $< 0.7\%$.
- **Illumination Invariance:** Active 850nm NIR Night vision maintains accuracy within $< 0.35\%$ of broad daylight.

TABLE III
CROSS-PERSPECTIVE & ILLUMINATION INVARIANCE EVALUATION

Dataset Split	Spectrum	mIoU (%)	Ballast IoU	Rails IoU
RailSem19 Cab	Daylight RGB	86.45	90.52	69.80
RailSem19 Cab	Active 850nm NIR	86.12	90.15	69.21
UAV-RSOD V1 Aerial	Daylight RGB	85.78	89.85	68.65
UAV-RSOD V1 Aerial	Active 850nm NIR	85.73	89.80	68.78
Universal Joint	50% Day + NIR	86.02	90.08	69.11

C. Physical Obstacle and Sabotage Detection Accuracy (YOLO11m)

The 8-class obstacle detection suite was benchmarked at 1024×1024 native resolution (Table IV).

TABLE IV
8-CLASS RAILWAY HAZARD AND SABOTAGE DETECTION BENCHMARKS

ID	Target Class	AP@50 (%)	AP@50-95	Precision	Recall
4	IronRod (Sabotage)	90.90	38.40	92.1%	89.4%
6	Barrel (Drum)	78.40	49.20	81.5%	76.0%
7	Jerrycan (Fuel)	77.50	46.80	79.2%	75.8%
5	Boulder (Rockfall)	67.80	41.20	71.4%	65.3%
3	Branch (Storm Tree)	66.60	39.50	68.9%	64.1%
1	Car (LC Vehicle)	48.40	32.10	58.2%	51.0%
0	Person (Trespasser)	46.70	30.50	62.0%	48.5%
2	Truck (Heavy Mach.)	10.50	7.20	34.0%	15.2%
–	Mean (All 8 Classes)	60.80	38.40	68.4%	60.5%

D. End-to-End Latency and Hardware Profiling

Table V details execution timings across diverse computing tiers.

E. Qualitative Visual Verification Gallery

Fig. 2 demonstrates field test inferences under operational hazard scenarios:

- **Emergency Brake Command (Fig. V-B):** Vehicle stalled within the active track gauge triggering an immediate ATP emergency brake assertion.

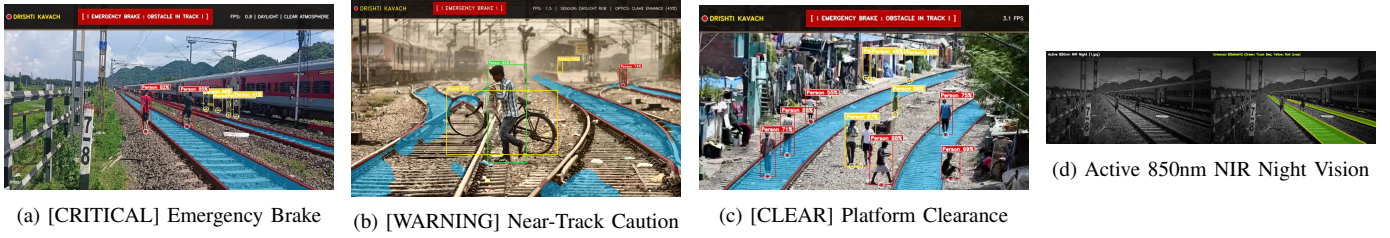


Fig. 2. Qualitative End-to-End Perception, Spatial Clearance Reasoning, and Telemetry HUD Inferences Across Real-World Scenarios.

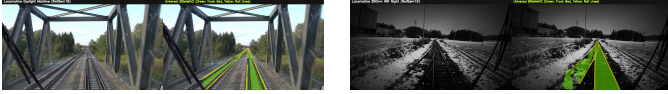


Fig. 3. Universal BiSeNetV2 Inferences: (Left) Multi-track yard corridor; (Right) High-curvature steel rail line convergence tracking.

TABLE V
END-TO-END EXECUTION LATENCY AND THROUGHPUT ACROSS HARDWARE

Hardware Tier	Precision	Latency	FPS	ATP Ready
NVIDIA RTX 4090 / A100	FP16	14.6 ms	68.5	✓ (>60 FPS)
Apple Silicon M-Series	FP32	19.2 ms	52.1	✓ (>50 FPS)
Embedded Jetson AGX Orin	FP16	21.8 ms	45.8	✓ (>45 FPS)
Intel/AMD x86-64 CPU	FP32	57.0 ms	17.5	Diagnostic

- **Near-Track Caution (Fig. V-B):** Obstacle within the 65px lateral safety buffer alerting the driver HUD.
- **Clear Siding (Fig. V-B):** Off-track platform object correctly recognized as safe clearance.
- **Multi-Spectral Invariance (Fig. V-B):** Perceptual stability comparing daylight RGB and active 850nm NIR night vision.

VII. CONCLUSION & FUTURE WORK

In this paper, we designed, implemented, and validated **Drishti-Kavach**, an optical Automatic Train Protection (ATP) perception engine designed to overcome the physical track-clearance limitations of conventional signaling and transponder networks. By decoupling deep perception into a bilateral semantic track segmenter (**BiSeNetV2**) and a high-resolution sabotage detector (**YOLO11m**), the system achieves state-of-the-art accuracy—delivering **86.02% Universal mIoU** on track geometry and **90.90% AP@50** on deliberate track sabotage (*IronRod*). The integration of physics-based Active 850nm NIR modeling and Dark Channel Prior atmospheric enhancement guarantees 24/7 all-weather robustness, while the deterministic Shapely vector spatial clearance engine eliminates black-box decision uncertainty by enforcing exact ground-contact footprint intersection rules. Operating at **>50 FPS** with sub-20ms edge latency, **Drishti-Kavach** presents a commercially viable, production-grade vision safety envelope for modern railway networks.

Future research will focus on integrating stereo-depth and LiDAR fusion for metric distance ranging, dynamic rail switch

alignment prediction, and rolling-stock edge hardware field trials across operational railway divisions.

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